Photometric Test Report

Reference Standards

ANSI C82.77-10-2014
IES LM-79-2019

Prepared For

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Test Laboratory:
UL-CCIC Company Limited
Test Laboratory Address:
No.2, Chengwan Road, Suzhou Industrial Park, Suzhou 215122, China
Catalog Number
BPANEL-50S54-UNV-D-8SC61-14-G-WH-P

Project Number
4791498147.3
Report Number
4791498147.3-1a

Test Date
2024-10-30~2024-11-05
Issue Date
2024-11-21
Revision Date
N/A

Prepared By	**Approved By**
Becky Fan	Roger Xu
Becky Fan	Roger Xu

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Test List

Sample Received Date: 2024-10-22

Test Item	Sample ID	Model Number	Tests Conducted By
Integrating Sphere Test	7733859	BPANEL-50S54-UNV-D-8SC61-14-G-WH-P	Hom Dong
Goniophotometer Test	7733859	BPANEL-50S54-UNV-D-8SC61-14-G-WH-P	Hom Dong
THD and PF Test	7733859	BPANEL-50S54-UNV-D-8SC61-14-G-WH-P	Hom Dong

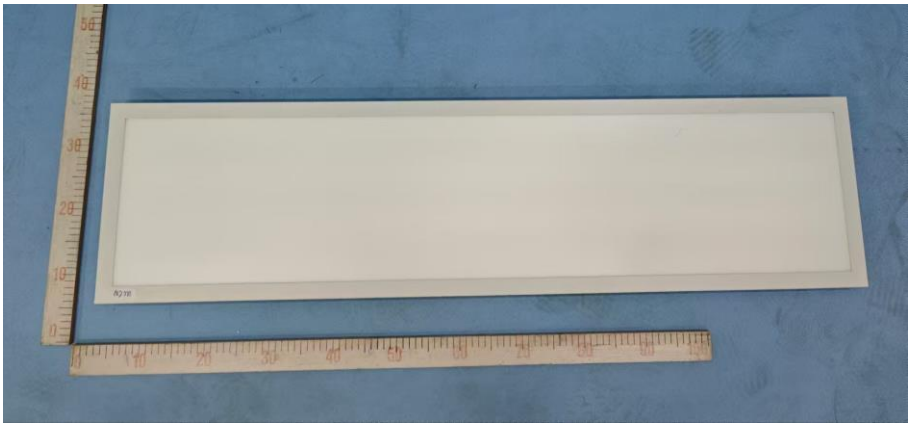
Remark (if any)

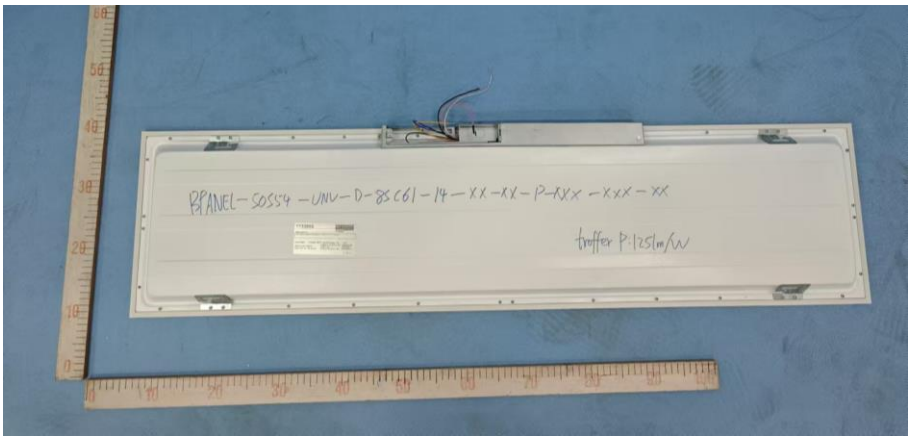
1. UL test equipment information is recorded on Meter Use in UL's Aurora database.

Product Description

Luminaire Description: 1x4 Luminaires for Ambient Lighting of Interior Commercial Spaces
Model Number: BPANEL-50S54-UNV-D-8SC61-14-G-WH-P
Electrical Ratings and CCT: 120-277Vac, 50/60 Hz, 50W max, 3000K/3500K/4000K/5000K/5700K/6500K tunable
Driver Model Number: BQ-50L6-1250-40-M56
LED Package: BXVN-30E-13H-9NV, BXVN-65E-13H-9NV, Bridgelux Inc
Family Model and Variation: BPANEL-50S54-UNV-D-8SC61-14-XX-YY-P-ZZZ-***-##, where "XX" represent mounting, can be G=Grid mounting, SF=Surface mounting, SP=Suspended mounting, DF=Drywall flange mounting. "YY" represent Finish color, can be WH=White or others. "ZZZ" represent Emergency, can be EM*= Emergency types(* can be blank or any letter), blank=No emergency. "***" represent Sensor receptacle, can be SR#= Sensor receptacle types(# can be blank or any number), blank= Without sensor receptacle. ## represent junction box, can be JB=with junction box, blank=without junction box.

Photos of Products Characteristics





Integrating Sphere Test

Model No.	BPANEL-50S54-UNV-D-8SC61-14-G-WH-P	Sample ID.	7733859
Operate time (Min.)	55	Stabilization time (Min.)	50

Test Method

1.The sample was tested according to the IES LM-79-2019, and the product is assume to be brand new without seasoning.
2.Photometric paramters were measured using an integrating sphere, a spectroradiometer and software. The ambient temperature condition inside the sphere was maintained at 25 °C ± 0.9 °C.The reference standard lamp is power 100W omni-directional Incandescent lamp and was calibrated by National Institute of Metrology, China.
3.The sample measurements were made using a spectroradiometer connected by a fiber optic cable and detector through the detector port of the integrating sphere. Coating reflectance of the integrating sphere was 90% to 98%. Photometric measurement conditions was using 4π geometry. The self-absorption factor is applied in the final test result. The sample was operated at rated voltage and was stabilized before measurement. Chromaticity coordinates, correlated color temperature and color rendering index were calculated from the spectral radiant flux measurements taken at 1 nm intervals over the range of 380 to 780 nm.

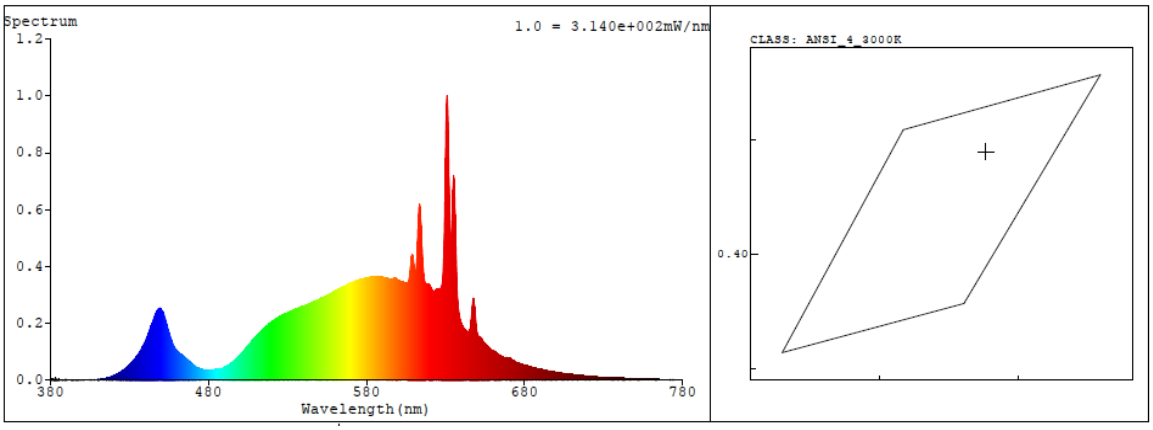
Integrating Sphere Test Conditions

Temperature (°C)	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Orientation
25.1	120	60	0.412	48.97	0.991	Horizontal

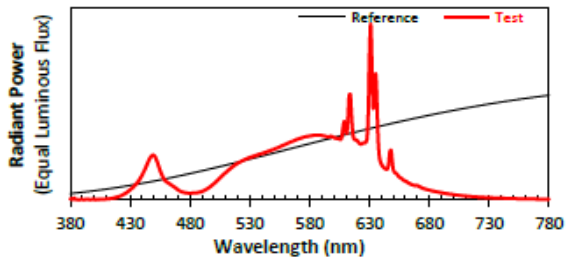
Test Results

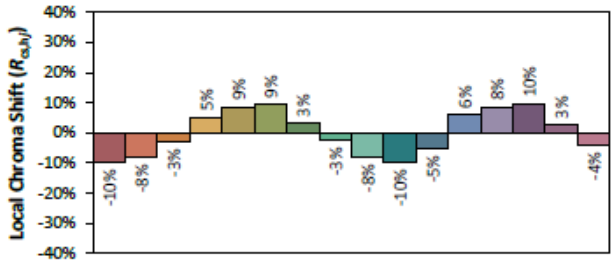
CCT (K)	CRI (Ra)	R9	Rf	Rg	Flux (lm)	Luminous Efficacy (lm/W)
3027	81	21	81	100	6430.7	131.3

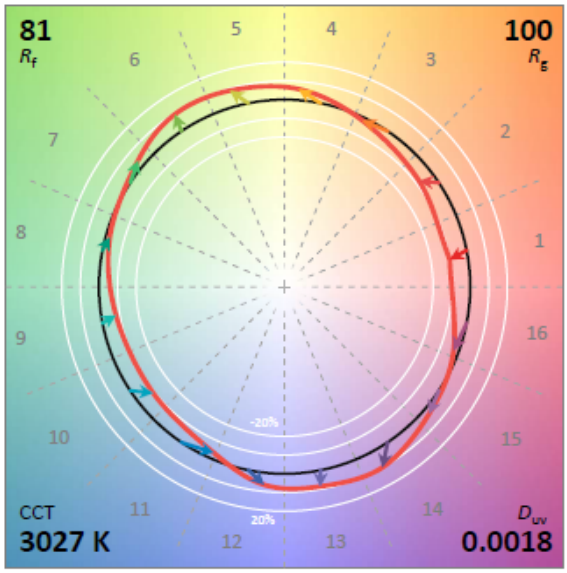
Spectrum

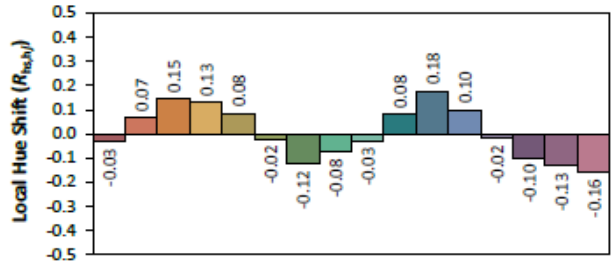


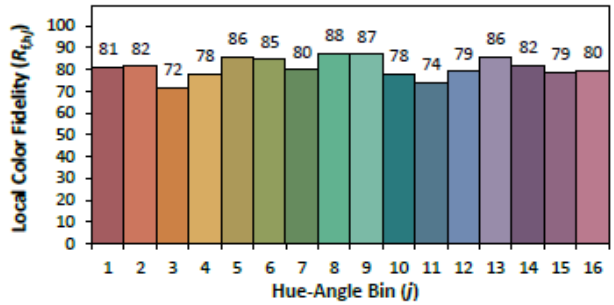
ANSI/IES TM-30-18 Color Rendition Report

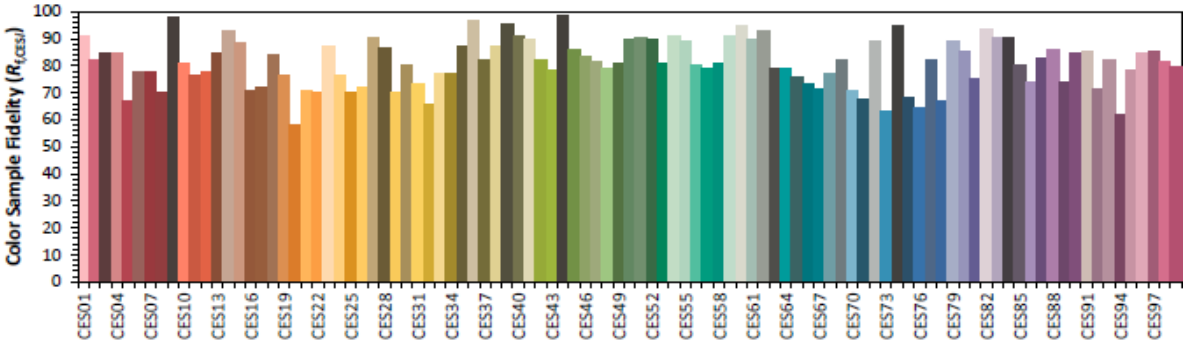












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.4377
y	0.4088
u'	0.2490
v'	0.5234

CIE 13.3-1995 (CRI)	
R_a	81
R_9	21

Colors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.00.

Integrating Sphere Test

Model No.	BPANEL-50S54-UNV-D-8SC61-14-G-WH-P	Sample ID.	7733859
Operate time (Min.)	55	Stabilization time (Min.)	50

Test Method

1.The sample was tested according to the IES LM-79-2019, and the product is assume to be brand new without seasoning.
2.Photometric paramters were measured using an integrating sphere, a spectroradiometer and software. The ambient temperature condition inside the sphere was maintained at 25 °C ± 0.9 °C.The reference standard lamp is power 100W omni-directional Incandescent lamp and was calibrated by National Institute of Metrology, China.
3.The sample measurements were made using a spectroradiometer connected by a fiber optic cable and detector through the detector port of the integrating sphere. Coating reflectance of the integrating sphere was 90% to 98%. Photometric measurement conditions was using 4π geometry. The self-absorption factor is applied in the final test result. The sample was operated at rated voltage and was stabilized before measurement. Chromaticity coordinates, correlated color temperature and color rendering index were calculated from the spectral radiant flux measurements taken at 1 nm intervals over the range of 380 to 780 nm.

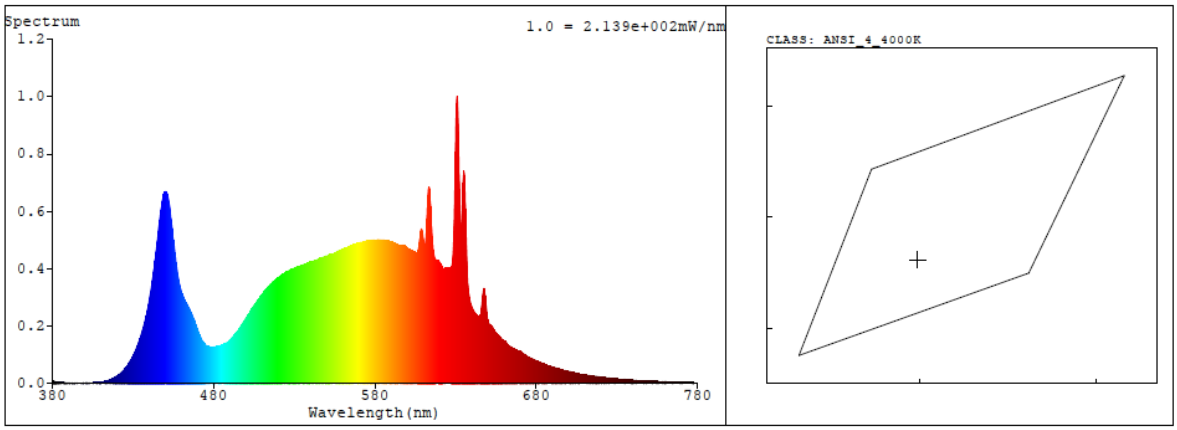
Integrating Sphere Test Conditions

Temperature (°C)	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Orientation
25.1	120	60	0.411	48.84	0.991	Horizontal

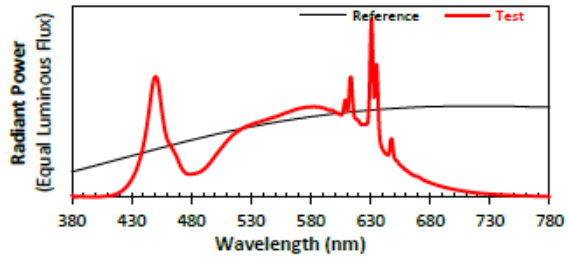
Test Results

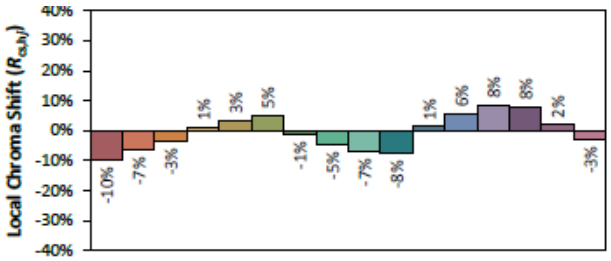
CCT (K)	CRI (Ra)	R9	Rf	Rg	Flux (lm)	Luminous Efficacy (lm/W)
4011	84	28	84	98	6526.5	133.6

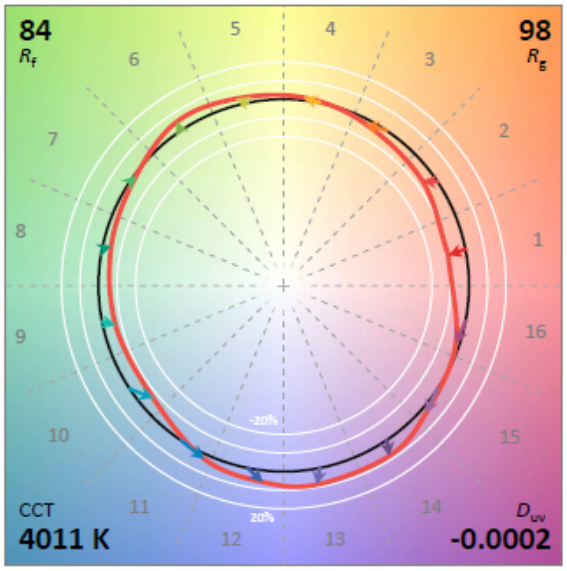
Spectrum

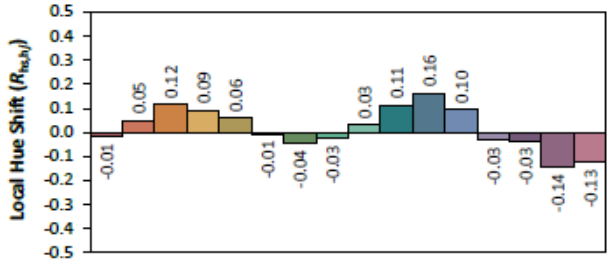


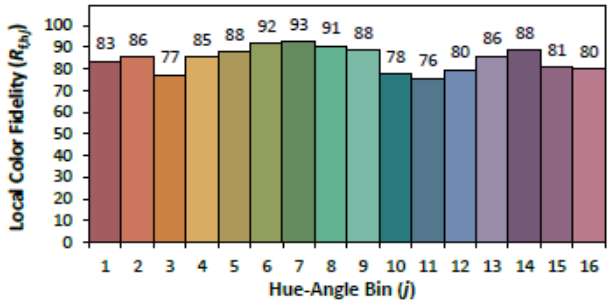
ANSI/IES TM-30-18 Color Rendition Report

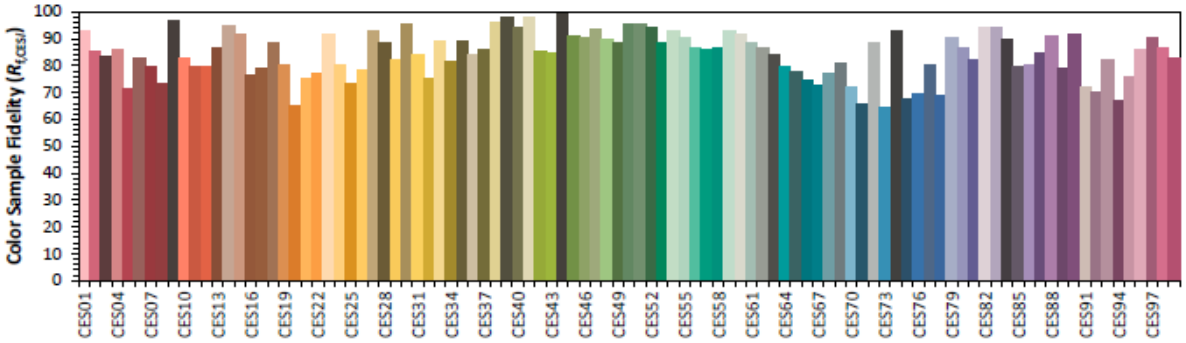












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.3799
y	0.3761
u'	0.2250
v'	0.5012

CIE 13.3-1995 (CRI)	
R_a	84
R_9	28

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Integrating Sphere Test

Model No.	BPANEL-50S54-UNV-D-8SC61-14-G-WH-P		Sample ID.	7733859
Operate time (Min.)	55		Stabilization time (Min.)	50

Test Method

1.The sample was tested according to the IES LM-79-2019, and the product is assume to be brand new without seasoning.
2.Photometric paramters were measured using an integrating sphere, a spectroradiometer and software. The ambient temperature condition inside the sphere was maintained at 25 °C ± 0.9 °C.The reference standard lamp is power 100W omni-directional Incandescent lamp and was calibrated by National Institute of Metrology, China.
3.The sample measurements were made using a spectroradiometer connected by a fiber optic cable and detector through the detector port of the integrating sphere. Coating reflectance of the integrating sphere was 90% to 98%. Photometric measurement conditions was using 4π geometry. The self-absorption factor is applied in the final test result. The sample was operated at rated voltage and was stabilized before measurement. Chromaticity coordinates, correlated color temperature and color rendering index were calculated from the spectral radiant flux measurements taken at 1 nm intervals over the range of 380 to 780 nm.

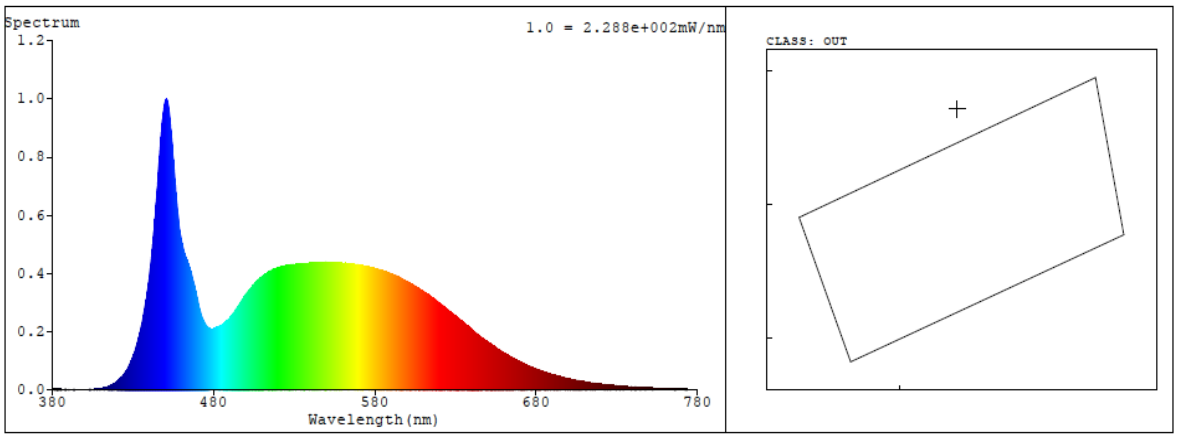
Integrating Sphere Test Conditions

Temperature (°C)	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Orientation
25.1	120	60	0.414	49.24	0.9912	Horizontal

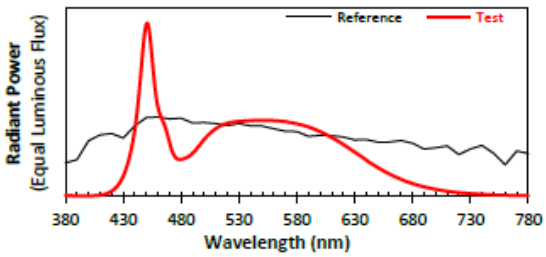
Test Results

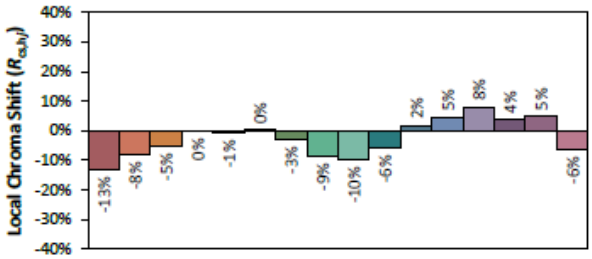
CCT (K)	CRI (Ra)	R9	Rf	Rg	Flux (lm)	Luminous Efficacy (lm/W)
6476	82	1	83	94	6530.7	132.6

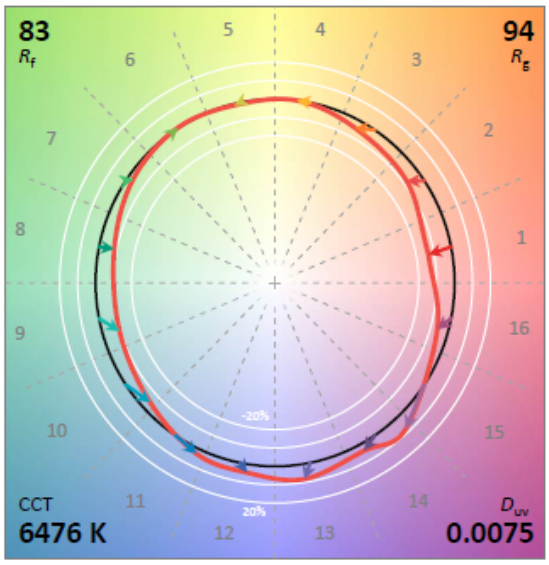
Spectrum

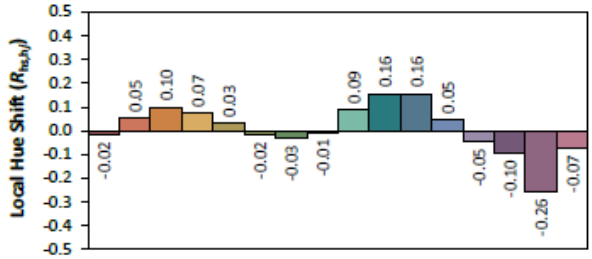


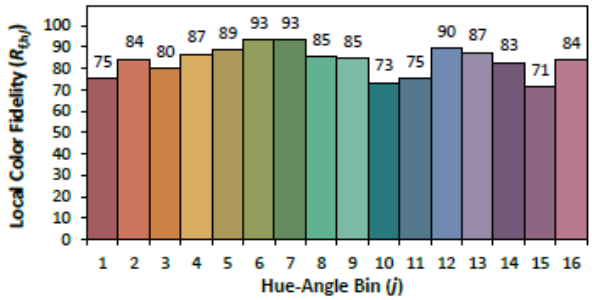
ANSI/IES TM-30-18 Color Rendition Report

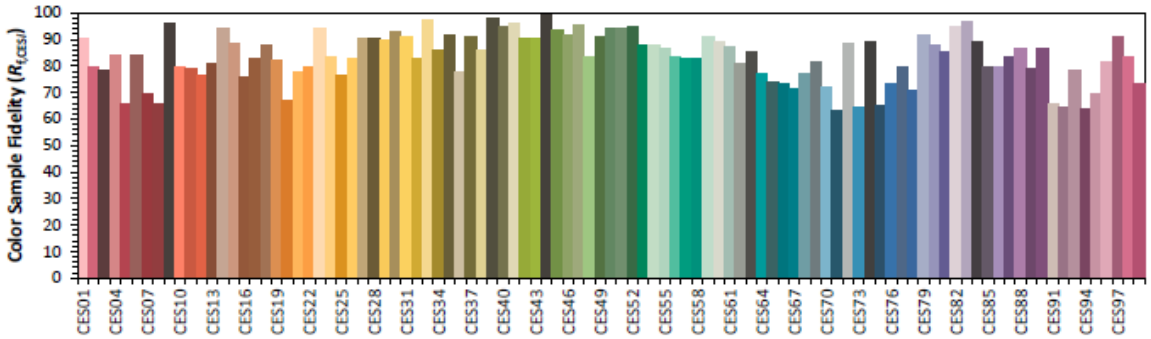












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.3121
y	0.3370
u'	0.1945
v'	0.4724

CIE 13.3-1995 (CRI)	
R_a	82
R_9	1

Colors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.00.

Goniophotometer Test

Model No.	BPANEL-50S54-UNV-D-8SC61-14-G-WH-P	Sample ID.	7733859
Operate time (Min.)	65	Stabilization time (Min.)	50

Test Method

1.The sample was tested according to the IES LM-79-2019, and the product is assume to be brand new without seasoning.
2.Photometric parameters were measured using a type C goniophotometer and software.
3.The ambient temperature shall be maintained at 25 °C ± 0.9 °C, measured at a point not more than 1.5 m from the sample and at the same height as the sample. The reference standard lamp is power 400W omni-directional Incandescent lamp and was calibrated by National Institute of Metrology, China.
4.The samples were operated at rated voltage and was stabilized before measurement. Luminous flux, luminaire efficacy, zonallumen were calculated from the software taken at 1° vertical intervals and 22.5° horizontal intervals. Photometric distance was more than five times of the largest dimension of the test SSL product.

Goniophotometer Test Conditions

Temperature (°C)	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Orientation
24.3	120	60	0.413	49.13	0.9925	Horizontal

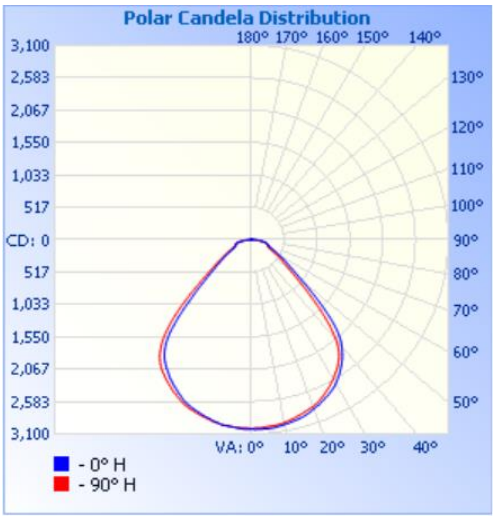
Test Results

Zonal Lumen Requirement (0°-60°)	Beam Angle (50%)		Spacing Criteria (0-180°)	Spacing Criteria (90°-270°)	Flux (lm)	Luminous Efficacy (lm/W)
	Horizontal Spread	Vertical Spread				
86.5%	90.6	90.9	1.30	1.28	6447	131.2

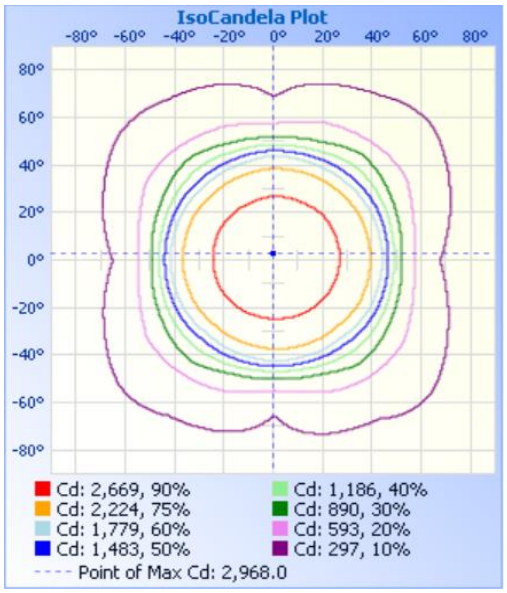
UGR
20.4

Goniophotometer Test (Cont'd)

Polar Candela Distribution



IsoCandela Plot



Goniophotometer Test (Cont'd)

Zonal Lumen Summary

Zonal Lumen Summary		
Zone	Lumens	% Luminaire
0-30	2,326.1	36.1%
0-40	3,787.6	58.7%
0-60	5,574.9	86.5%
60-90	855.2	13.3%
70-100	430.4	6.7%
90-120	6.3	0.1%
0-90	6,430.0	99.7%
90-180	17.0	0.3%
0-180	6,447.0	100%

Lumens Per Zone

Lumens Per Zone					
Zone	Lumens	% Total	Zone	Lumens	% Total
0-10	280.6	4.4%	90-100	3.6	0.1%
10-20	808.6	12.5%	100-110	1.3	0%
20-30	1,236.8	19.2%	110-120	1.4	0%
30-40	1,461.5	22.7%	120-130	1.7	0%
40-50	1,136.9	17.6%	130-140	2.1	0%
50-60	650.4	10.1%	140-150	2.3	0%
60-70	428.3	6.6%	150-160	2.1	0%
70-80	297.0	4.6%	160-170	1.6	0%
80-90	129.8	2.0%	170-180	0.7	0%

Goniophotometer Test (Cont'd)

Intensity Data(cd)

	0	22.5	45	67.5	90	112.5	135	157.5	180	202.5	225	247.5	270	292.5	315	337.5	360
0	2958	2958	2958	2958	2958	2958	2958	2958	2958	2958	2958	2958	2958	2958	2958	2958	2958
1	**2968**	2963	2956	2950	2945	2946	2957	2962	2965	2961	2954	2942	2946	2951	2958	2963	2963
2	2966	2962	2954	2943	2944	2951	2961	2961	2964	2962	2952	2945	2946	2948	2962	2961	2964
3	2965	2959	2955	2944	2943	2948	2960	2958	2959	2956	2950	2943	2947	2952	2965	2967	2966
4	2964	2960	2953	2942	2936	2945	2956	2954	2951	2955	2946	2939	2946	2952	2961	2965	2967
5	2965	2954	2947	2938	2930	2940	2950	2949	2947	2949	2941	2937	2942	2945	2957	2962	2961
6	2958	2955	2946	2934	2929	2933	2944	2943	2940	2945	2939	2940	2940	2945	2957	2956	2957
7	2959	2947	2941	2933	2927	2924	2935	2944	2939	2943	2939	2931	2932	2943	2957	2957	2949
8	2950	2944	2937	2925	2921	2926	2935	2931	2934	2932	2933	2929	2931	2938	2948	2957	2948
9	2943	2940	2928	2920	2910	2914	2928	2932	2929	2929	2926	2927	2930	2932	2949	2949	2947
10	2932	2930	2923	2914	2908	2909	2912	2917	2914	2920	2912	2916	2917	2925	2940	2945	2936
11	2931	2924	2915	2904	2894	2892	2903	2907	2904	2904	2904	2904	2909	2921	2936	2931	2926
12	2915	2907	2900	2891	2879	2882	2889	2891	2890	2891	2891	2895	2902	2904	2917	2923	2920
13	2908	2902	2886	2879	2872	2867	2874	2872	2872	2882	2882	2880	2887	2893	2911	2910	2904
14	2900	2885	2878	2860	2852	2856	2858	2865	2861	2863	2866	2871	2873	2886	2897	2896	2897
15	2889	2875	2860	2851	2842	2835	2845	2844	2849	2854	2855	2858	2868	2872	2887	2884	2881
16	2882	2860	2851	2835	2829	2826	2835	2831	2830	2835	2846	2843	2853	2863	2871	2876	2872
17	2864	2850	2833	2826	2813	2808	2821	2811	2816	2822	2831	2832	2844	2849	2861	2861	2861
18	2852	2835	2827	2807	2801	2795	2806	2802	2801	2810	2818	2816	2831	2841	2849	2853	2848
19	2833	2820	2808	2799	2785	2782	2794	2782	2789	2797	2809	2811	2818	2825	2836	2835	2834
20	2814	2809	2799	2781	2777	2766	2773	2767	2771	2779	2786	2793	2809	2817	2829	2826	2823
25	2719	2706	2691	2674	2664	2653	2649	2649	2651	2662	2669	2690	2709	2719	2727	2721	2716
30	2599	2562	2542	2529	2521	2499	2496	2490	2508	2517	2540	2555	2579	2590	2592	2588	2587
35	2415	2382	2336	2337	2335	2297	2279	2298	2329	2342	2357	2391	2434	2437	2424	2432	2431
40	2168	2045	1940	1975	2022	1916	1779	1897	2021	1973	1918	2085	2173	2145	2100	2137	2154
45	1702	1493	1410	1404	1402	1351	1309	1332	1369	1387	1400	1524	1663	1606	1552	1594	1614
50	1085	1047	1066	943	862	927	999	877	855	910	1065	1042	1039	1115	1173	1099	1007
55	716	719	778	692	585	682	719	644	577	655	793	734	676	768	878	758	678
60	522	568	525	572	410	559	485	543	407	544	547	585	476	610	607	604	491
65	368	498	392	482	301	476	379	467	298	474	398	502	336	527	419	511	360
70	288	421	347	381	264	375	323	364	261	385	339	414	270	443	360	422	290
75	253	311	288	280	245	277	276	264	243	276	278	309	249	330	301	316	252
80	206	215	236	196	180	199	221	187	178	198	236	217	204	220	249	217	201
85	130	133	146	111	88	100	119	96	86	113	164	134	120	140	175	138	120
90	43	24	13	6	2	2	5	2	2	6	33	22	28	36	42	38	35
95	2	2	2	2	1	2	1	2	2	2	2	2	1	2	2	3	2
100	2	1	1	2	1	2	1	1	2	1	1	1	1	1	2	2	2
105	1	1	2	1	1	1	1	1	1	1	1	1	1	2	1	1	1
110	1	1	1	1	1	1	1	1	1	1	1	1	2	1	1	1	1
115	1	2	1	2	1	2	2	2	1	2	1	2	1	1	2	1	1
120	2	1	2	2	2	2	2	2	2	1	1	2	1	2	2	2	2
125	1	2	2	2	2	2	2	2	2	2	2	2	2	2	1	2	2
130	2	3	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
135	3	3	3	3	3	3	3	3	2	3	2	3	2	2	3	3	2
140	3	3	4	3	4	4	4	4	3	4	3	4	3	3	3	3	3
145	3	4	3	4	4	4	4	4	4	4	4	4	4	4	3	4	4
150	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
155	4	4	4	5	4	5	5	5	4	5	5	5	5	5	4	5	5
160	5	5	5	6	5	6	5	6	5	5	5	5	5	5	5	5	5
165	6	6	6	6	6	5	5	6	6	6	6	6	6	5	5	6	6
170	7	7	7	7	6	6	6	7	7	8	7	7	6	6	6	7	7
175	9	8	8	7	7	7	7	8	8	8	7	7	7	6	7	8	8
180	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7

Corrected UGR values generated per CIE 190-2010

UGR Table - Corrected

Reflectances											
Ceiling Cavity		70	70	50	50	30	70	70	50	50	30
Walls		50	30	50	30	30	50	30	50	30	30
Floor Cavity		20	20	20	20	20	20	20	20	20	20
Room Size		UGR Viewed Crosswise					UGR Viewed Endwise				
X=2H	Y=2H	15.5	17.0	15.9	17.3	17.6	15.0	16.5	15.4	16.8	17.1
	3H	17.1	18.5	17.5	18.8	19.1	16.6	17.9	17.0	18.2	18.6
	4H	17.9	19.1	18.3	19.5	19.9	17.4	18.6	17.8	18.9	19.3
	6H	18.7	19.8	19.1	20.2	20.6	18.1	19.2	18.5	19.6	20.0
	8H	19.0	20.1	19.4	20.5	20.9	18.4	19.5	18.8	19.9	20.3
	12H	19.3	20.4	19.8	20.7	21.2	18.6	19.6	19.0	20.0	20.4
4H	2H	15.9	17.1	16.3	17.5	17.9	15.4	16.7	15.8	17.0	17.4
	3H	18.0	19.0	18.4	19.4	19.8	17.4	18.5	17.8	18.9	19.3
	4H	19.0	19.9	19.5	20.4	20.8	18.4	19.4	18.9	19.8	20.2
	6H	20.0	20.8	20.4	21.2	21.7	19.4	20.2	19.8	20.6	21.1
	8H	20.4	21.1	20.9	21.6	22.1	19.7	20.5	20.2	20.9	21.4
	12H	20.8	21.5	21.3	21.9	22.4	20.0	20.7	20.5	21.2	21.6
8H	4H	19.5	20.2	19.9	20.7	21.1	18.9	19.7	19.4	20.1	20.6
	6H	20.6	21.3	21.1	21.7	22.2	20.1	20.7	20.6	21.2	21.7
	8H	21.2	21.8	21.7	22.3	22.8	20.5	21.1	21.1	21.6	22.1
	12H	21.7	22.2	22.2	22.7	23.3	20.9	21.4	21.4	21.9	22.5
12H	4H	19.5	20.2	20.0	20.7	21.2	19.0	19.7	19.5	20.2	20.6
	6H	20.8	21.4	21.3	21.8	22.4	20.2	20.8	20.8	21.3	21.8
	8H	21.4	22.0	22.0	22.4	23.0	20.8	21.3	21.3	21.8	22.4

Maximum UGR = 23.3

THD and PF Test

Model No.	BPANEL-50S54-UNV-D-8SC61-14-G-WH-P	Sample ID.	7733859

Test Method

1. The samples were tested according to the ANSI C82.77-10-2014.
2. The ambient temperature condition was maintained at 25 °C ± 1 °C. The sample measurement was made using a digital power meter and power supply. The sample was operated at rated voltage and stabilized before measurement. The total harmonic distortion were calculated from the digital power meter.

Test Results

CCT Range	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Current THD	Orientation
0%-3000K	120	60	0.412	49.1	0.9925	11.33%	Horizontal
	277	60	0.180	48.0	0.9622	9.88%	Horizontal
20%-3500K	120	60	0.410	48.9	0.9925	10.57%	Horizontal
	277	60	0.180	47.9	0.962	9.83%	Horizontal
40%-4000K	120	60	0.411	49.0	0.9925	10.50%	Horizontal
	277	60	0.180	47.9	0.9620	9.87%	Horizontal
60%-5000K	120	60	0.411	49.0	0.9925	10.56%	Horizontal
	277	60	0.180	48.0	0.9622	9.85%	Horizontal
80%-5700K	120	60	0.412	49.1	0.9925	10.55%	Horizontal
	277	60	0.180	48.0	0.9622	9.87%	Horizontal
100%-6500K	120	60	0.415	49.4	0.9926	10.48%	Horizontal
	277	60	0.181	48.3	0.9628	9.73%	Horizontal

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